

WID3013 Group Assignment

Practical Computer Vision Skill for Non-Technical Users

Group size: 5 students | Final demo: Hermes/OpenClaw or equivalent interface

1. Assignment Objective

In this group assignment, you will design and demonstrate a practical Hermes/OpenClaw skill that uses computer vision or image processing to help non-technical users complete a real task.

The target users are students or future professionals from fields such as:

- marketing
- business
- accounting
- music
- language
- history
- political science
- law
- education
- communication
- event management

Your project should not be a toy demo. It must help a real user complete a useful task.

```
visual input / visual data -> computer vision processing -> useful output -> better  
decision or productivity
```

2. What I Expect

Your skill must show that you understand computer vision concepts, but the final application must be useful for non-CS users.

A good project should answer:

1. Who is the user?
2. What real task are you helping?
3. What visual input is involved?
4. What image processing or computer vision technique is used?
5. What useful output is produced?
6. How does the output help the user make a decision or save time?

3. Important Rule

This is not only a chatbot assignment. Your project must include a meaningful computer vision or image-processing component.

Acceptable CV components include:

- OCR
- image enhancement
- image quality assessment
- object detection
- face or person detection with ethical boundaries
- pose estimation
- layout analysis
- document image processing
- chart generation from visual or tabular data
- image segmentation
- color analysis
- visual similarity search
- video frame analysis
- dashboard visualization
- image-to-report conversion

The CV component does not need to be a newly trained model. You may use existing libraries, pretrained models, APIs, or image-processing algorithms, but you must understand and explain them.

4. Suggested Project Topics

Topic 1: Marketing Campaign Visual Audit Skill

Target users	Marketing, business, communication students
Input	poster image, advertisement, product photo, Instagram post, event banner
CV component	OCR to extract poster text; color and contrast analysis; layout analysis; image quality assessment; optional object detection
Useful output	campaign audit report; readability score; call-to-action analysis; improvement suggestions; platform suitability
Example use	Upload a university event poster and ask the skill to audit it for Instagram.

Topic 2: Accounting Receipt / Invoice Assistant

Target users	Accounting, finance, business admin students
Input	receipt image, invoice image, expense claim document
CV component	OCR; document layout analysis; table extraction; image preprocessing for readability; blur detection
Useful output	merchant name; date; total amount; tax/SST if visible; expense category; CSV-ready claim table; blur or missing-information warning
Example use	Upload several receipts and generate an expense claim summary.

Topic 3: Legal Evidence Organizer Skill

Target users	Law, political science, journalism students
Input	screenshots, scanned documents, accident photos, WhatsApp screenshots, public notice images
CV component	OCR; screenshot classification; document image processing; image captioning if available; visible date extraction
Useful output	evidence log; visible text summary; detected dates or names; item classification; missing information checklist; case appendix draft
Example use	Upload screenshots and generate a structured evidence table.

Important boundary: The system must not decide guilt or truth. It only organizes visible information.

Topic 4: History / Museum Artifact Cataloguing Skill

Target users	History, museum studies, education students
Input	artifact photo, old document image, painting, monument photo, historical object
CV component	image captioning; OCR for inscriptions; object or motif detection; image quality assessment; visual similarity search if available
Useful output	artifact catalogue draft; visible feature description; metadata fields; teaching note; quiz questions; uncertainty statement
Example use	Upload an artifact image and generate a museum catalogue entry.

Topic 5: Language and Signage Localization Skill

Target users	Language, translation, communication students
Input	poster, menu, signboard, instruction label, bilingual notice, screenshot
CV component	OCR; text region detection; layout analysis; image preprocessing; optional translation pipeline
Useful output	extracted text; translated version; tone/register recommendation; layout warning if translated text is too long; localization report
Example use	Upload a menu or poster and check whether the Malay or English translation fits the visual layout.

Topic 6: Music Performance Content Review Skill

Target users	Music, performing arts, event promotion students
Input	performance video frame, concert poster, stage photo, short video sample
CV component	pose estimation; face/person detection; image quality assessment; brightness and blur analysis; thumbnail selection
Useful output	stage visibility report; thumbnail recommendation; lighting feedback; framing feedback; social media posting suggestion
Example use	Upload a performance video frame and ask which frame is best for a YouTube or Instagram thumbnail.

Important boundary: The system should comment on visual presentation, not personal attractiveness.

Topic 7: Political Campaign Visual Framing Analyzer

Target users	Political science, communication, media studies students
Input	campaign poster, news screenshot, public policy infographic, political advertisement
CV component	OCR; color analysis; layout analysis; symbol detection; image captioning if available
Useful output	visual framing report; slogan extraction; symbol summary; emotional framing analysis; neutrality warning; media analysis paragraph
Example use	Upload a campaign poster and analyze its visual framing without endorsing any political party.

Important boundary: The system must remain neutral and analyze communication strategy only.

Topic 8: Event Attendance and Feedback Dashboard Skill

Target users	Event managers, student societies, faculty admin, music clubs
Input	registration table, attendance data, feedback scores, event photos if applicable
CV component	chart generation; dashboard visualization; optional QR attendance image processing; optional OCR from attendance sheets
Useful output	attendance rate; cost per attendee; feedback trend; generated chart in Telegram; event improvement recommendation
Example use	Paste event attendance data and generate charts directly in Telegram.

Topic 9: Survey Dashboard for Social Science

Target users	Political science, education, communication, social science students
Input	survey result table, poll screenshot, CSV data, Google Form summary screenshot
CV component	chart generation; OCR if using screenshot input; bar chart or pie chart creation; visual dashboard design
Useful output	response distribution; majority/minority opinion; generated chart; report-ready interpretation; policy recommendation
Example use	Paste survey data and generate a public opinion dashboard with interpretation.

5. What Counts as Computer Vision Understanding?

Your project should clearly demonstrate at least two of the following CV ideas:

Image acquisition and preprocessing

- resizing
- cropping
- grayscale conversion
- denoising
- thresholding
- contrast enhancement
- blur detection
- brightness correction

Feature extraction

- color histogram
- edge detection
- shape features
- text region detection
- layout features
- keypoint or pose features

Recognition or detection

- OCR
- object detection
- logo detection
- person detection
- document type classification
- QR code detection

Segmentation or localization

- separating text from background
- detecting document regions
- identifying product area
- locating chart components
- isolating foreground objects

Visual reasoning and output

- turning OCR into a report
- turning image features into a recommendation
- turning data into a chart
- generating a dashboard image
- creating a structured table from a document image

6. Minimum Technical Requirements

A. Skill file

Your Hermes/OpenClaw skill file must include:

7. skill name
8. target user
9. real-world problem
10. input format
11. CV or image-processing method
12. step-by-step workflow
13. output format
14. limitation handling
15. ethical boundary

B. Working demo

Your group must demonstrate the skill in Hermes, OpenClaw, Telegram, or a simple local interface. The demo must show:

```
input -> processing -> useful output  
receipt image -> OCR -> expense table  
campaign data -> chart generation -> dashboard insight  
poster image -> OCR/layout analysis -> campaign audit
```

C. Visual output

Your system must produce at least one visual output, such as:

- chart image
- dashboard image
- annotated image
- extracted text table
- evidence table
- visual audit report
- before/after image enhancement
- document layout visualization

For Telegram-based projects, the chart or image should be sent directly in Telegram where possible.

D. Explanation of CV method

You must explain the method used. For example:

```
We used OCR to extract text from the poster. OCR converts text regions in an image into machine-readable text. We also used contrast analysis to check whether the text is readable against the background.
```

```
We used a bar chart to visualize cost per lead because marketing managers need to compare platform efficiency quickly.
```

7. Group Roles

Each group has 5 members. Suggested roles:

Member 1: Domain and User Research Lead

- identifying the target user
- explaining the real-world job task
- collecting realistic sample inputs
- checking whether the output is useful

Member 2: CV / Image Processing Lead

- OCR, preprocessing, detection, analysis, or chart generation
- explaining the technical method
- validating results

Member 3: Skill Workflow Designer

- Hermes/OpenClaw skill file
- prompt workflow
- input/output format
- error handling

Member 4: Tool / Integration Developer

- Telegram integration

- MCP/tool calling
- file handling
- chart/image output

Member 5: Evaluation and Presentation Lead

- testing with examples
- preparing slides
- explaining limitations
- running live demo

8. Presentation Format

Each group will present for 8 to 10 minutes.

- **Slide 1: Project Title** - Example: Receipt-to-Claim Assistant for Accounting Students
- **Slide 2: Target User and Problem** - Explain who the user is, what task they perform, and why the task is painful or time-consuming.
- **Slide 3: Why Computer Vision Is Needed** - Explain the visual input and why ordinary text processing is not enough.
- **Slide 4: System Workflow** - Show the pipeline from input image/data to CV method, structured information, skill output, and useful recommendation.
- **Slide 5: CV Method Used** - Explain what method was used, why it was suitable, and what its limitations are.
- **Slide 6: Skill Design** - Show your skill instruction structure: role, input, workflow, output format, and safety rules.
- **Slide 7: Demo Input** - Show your sample poster, receipt, screenshot, artifact photo, survey table, or campaign data.
- **Slide 8: Demo Output** - Show the chart, report, extracted table, annotated result, or recommendation.
- **Slide 9: Live Demo** - Run the skill.
- **Slide 10: Limitations and Future Work** - Examples: OCR fails on blurry images; small text is hard to read; chart only supports limited formats; no multimodal model available; future work can support Excel upload or full PDF report.

9. Assessment Rubric

Component	Marks
Real-world usefulness for art-stream users	15
Clear target user and job task	10
Correct use of computer vision / image processing concept	20
Skill workflow design and prompt quality	15
Working demo and system integration	15
Quality of output: report, chart, table, or visual result	10
Insightfulness and practical recommendation	10
Limitation and ethical awareness	5
Total	100

10. What I Do Not Want

Do not submit a project that is only:

- a normal chatbot with no CV component
- a generic dashboard with no meaningful interpretation
- a toy classifier unrelated to real users
- a cat/dog classifier
- a face attractiveness detector
- a race/gender/religion detector
- a detect-emotion-from-face system with no ethical discussion
- a poster analyzer that cannot process any image or data
- a chart generator that gives no decision-making insight

11. What Makes a Strong Project?

A strong project has this structure:

```
Real user -> real visual/data problem -> suitable CV technique -> structured output -> useful decision
```

```
Accounting assistant -> receipt images are difficult to enter manually -> OCR + document layout processing -> expense table and warning flags -> faster claim preparation
```

```
Marketing executive -> campaign performance is hard to compare -> visualization + KPI calculation -> chart and platform recommendation -> better budget allocation
```

```
Law student -> evidence screenshots are messy -> OCR + screenshot classification -> evidence log -> easier case report preparation
```

12. Final Message to Students

This assignment is not about making the most complicated AI system. It is about designing a useful computer vision skill.

A good skill should help a non-technical person do real work better.

The final output should not only say:

```
I detected something.
```

It should help the user answer:

```
What should I do next?
```

That is the standard expected for this assignment.